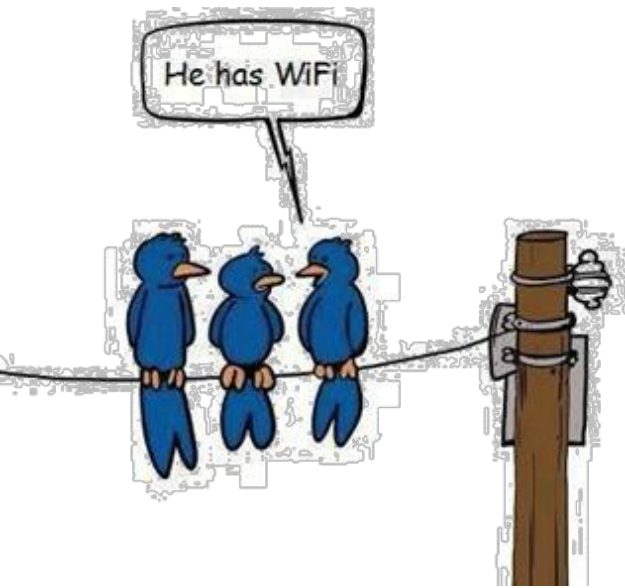


ECSE 4560/ 6560

Modern Communication Systems

Lecture 13: OFDM SFO



Ish Jain, Fall 2025



Announcements

- Join Piazza: <https://piazza.com/rpi/fall2025/ecse45606560>
- Per-Class Quiz
 - Posted on LMS; Due Today 11:59 PM
- Attendance Link: <https://forms.office.com/r/p6P9YrN1d8>
 - Today's Magic Word: **SFO**
- Projects
 - Group formation [MCS Fall25 project.xlsx](#)
- HW4 will be posted soon.
- Exam 1: Friday Oct 24
- Exam 2: Friday Nov 21
- Final Project Presentation and Demos: Dec 9 and Dec 12
- Project Report and peer evaluation: Finals exam week

Extra Reading on Piazza

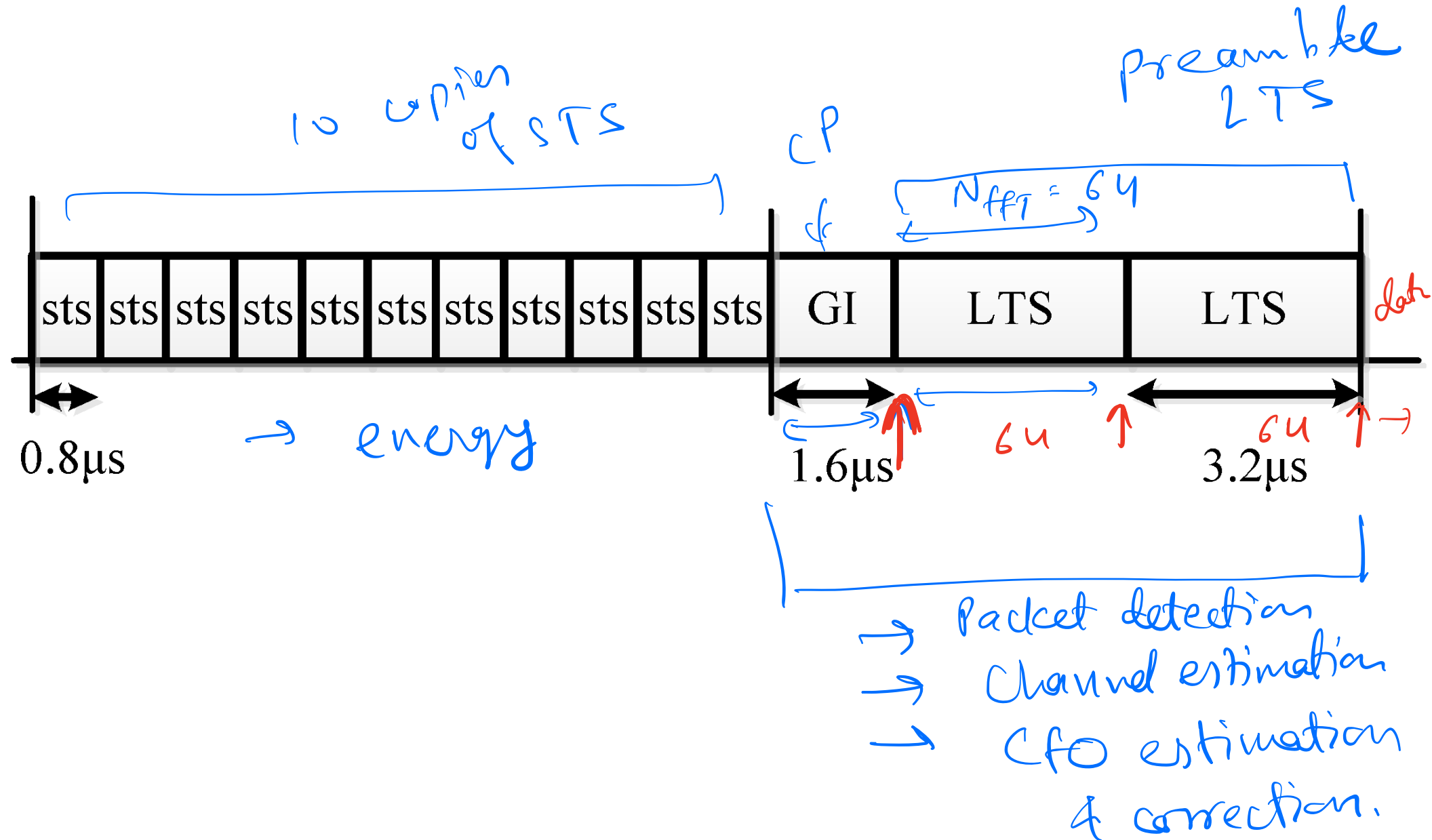
- Principles of Modern Wireless Communication Systems Theory and Practice; by Aditya K. Jagannatham, IIT Kanpur Book
 - Parts of Chapter 1,2,3,4,7
- An Adaptive Orthogonal Frequency Division Multiplexing Baseband Modem for Wideband Wireless Channels; by Jit Ken Tan, MIT Thesis
 - OFDM Decoding and Synchronization
 - Lab 4 Implementation

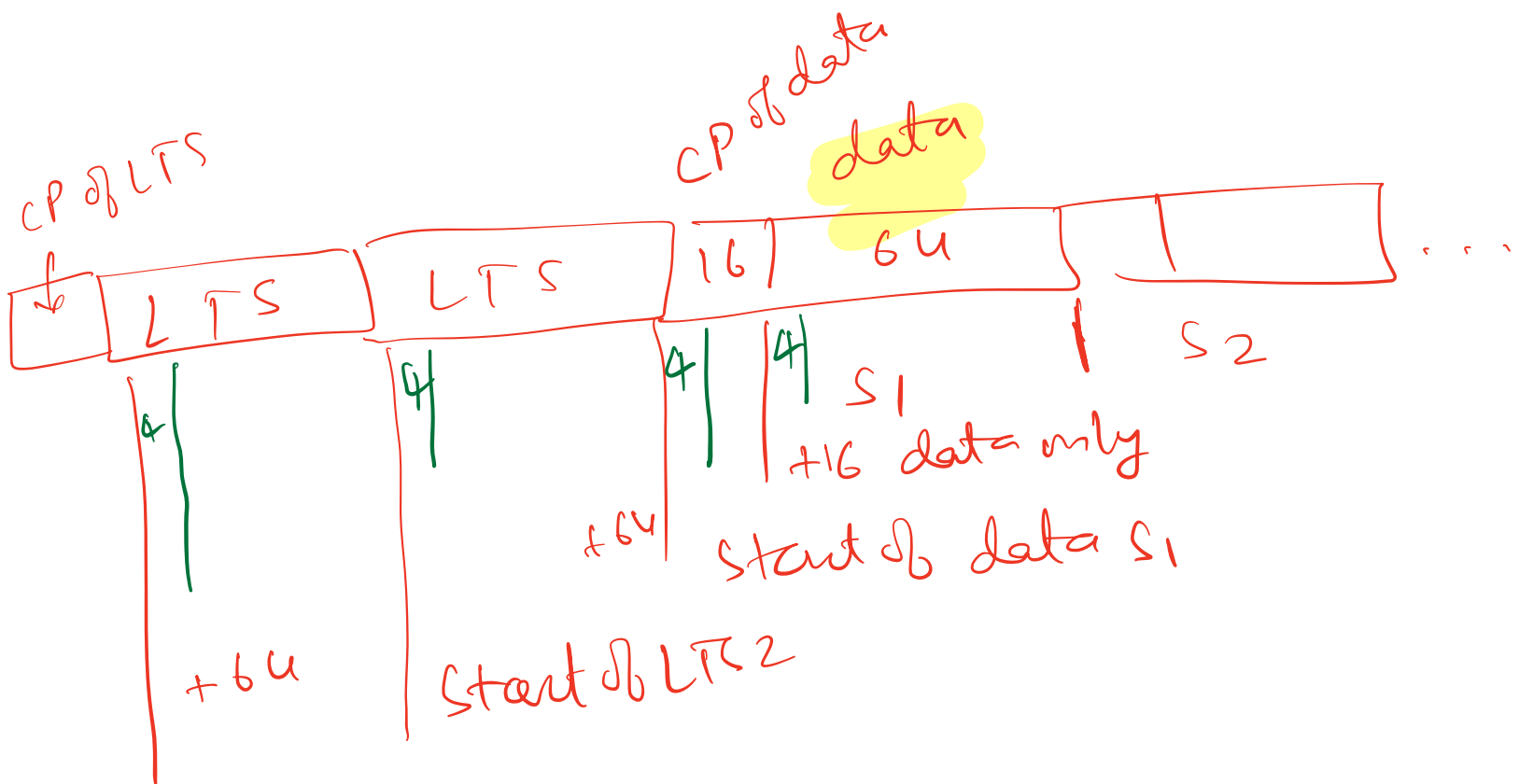
Today's Class: OFDM SFO

- OFDM Synchronization
 - CFO correction (Revise)
 - Channel Estimation (Revise)
 - SFO and Residual Phase correction
- 

WiFi Example: STS and LTS

HW 4

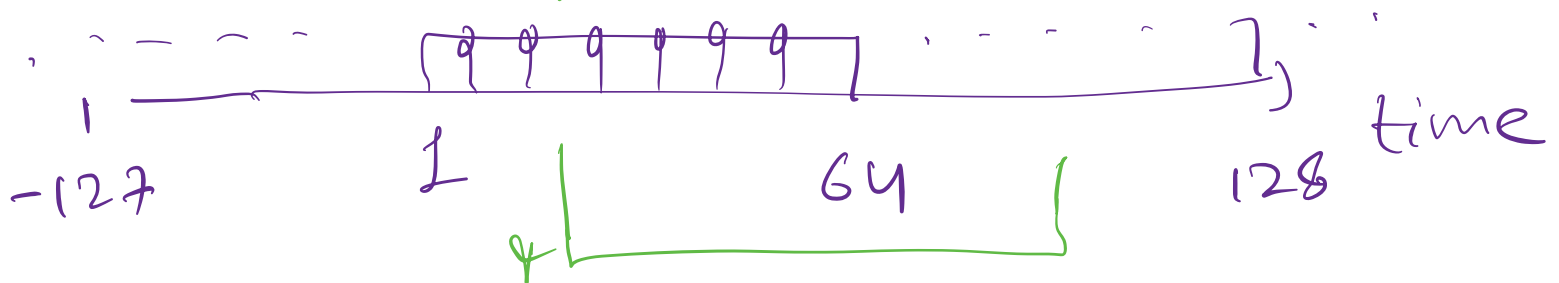




Start of LTS 1

$\Rightarrow$ given by packet detection algo
 $\rightarrow$ cross-correlation

CP added front or back of data?

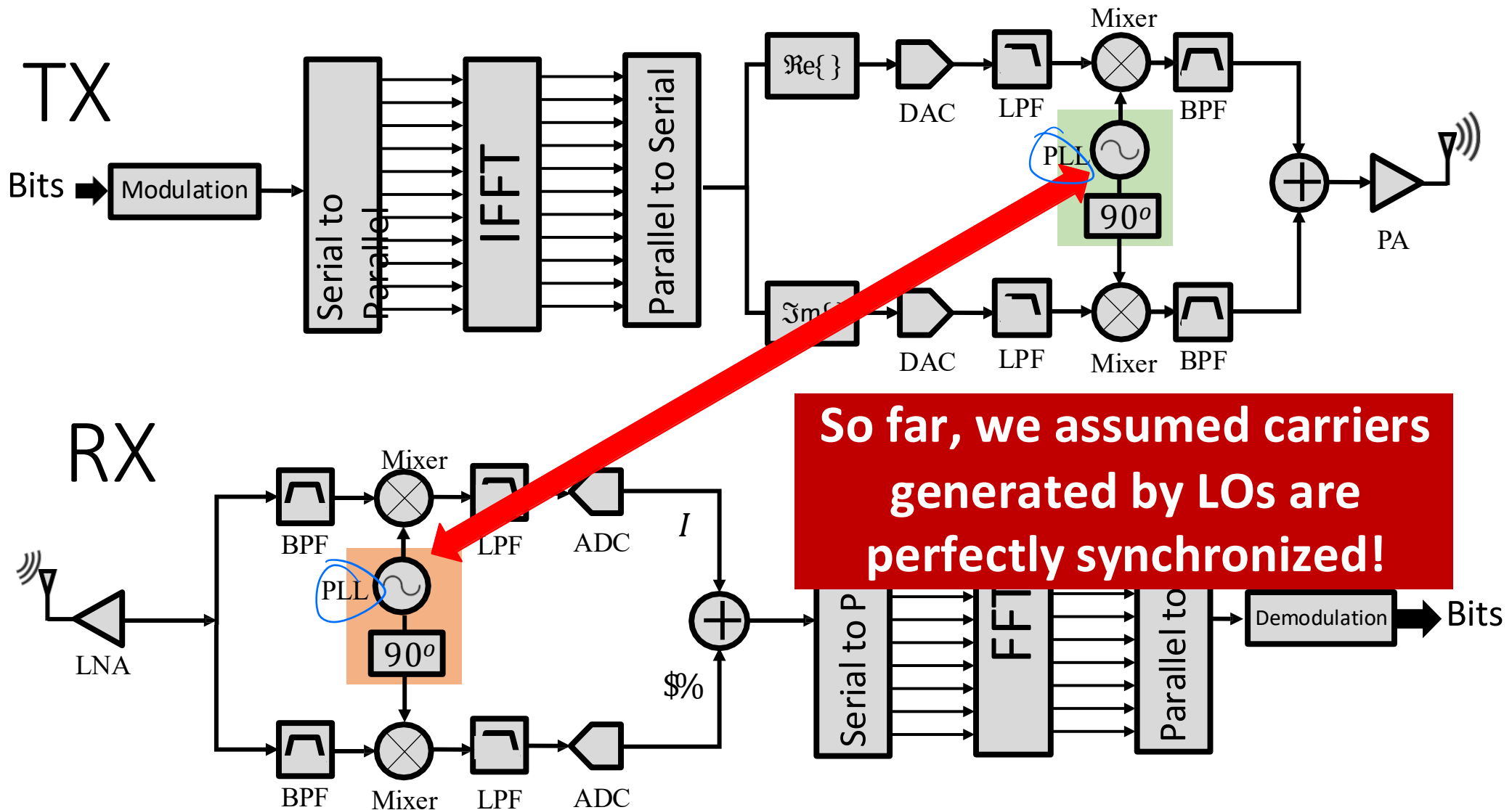


$$x(t) \rightarrow x(k) \quad \downarrow \text{FFT} \quad \rightarrow \sum_{k=0}^{64+4-1} x(4T_s) e^{-j2\pi k(4T_s)}$$

$$x(t-4) \rightarrow x(k) e$$

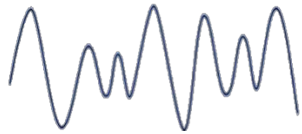
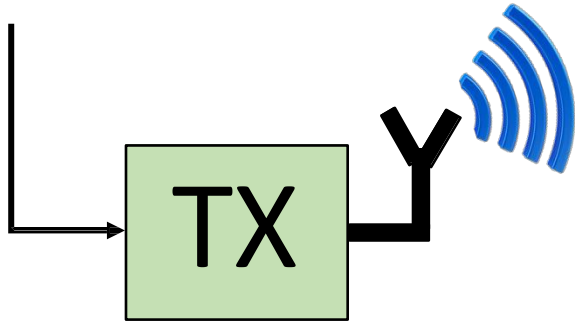
OFDM: Orthogonal Frequency Division

Multiplexing Transmit Symbols in Frequency Domain On Orthogonal Subcarriers

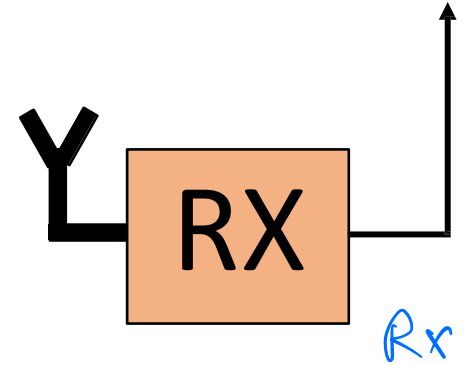


Carrier Frequency Offset

101101011



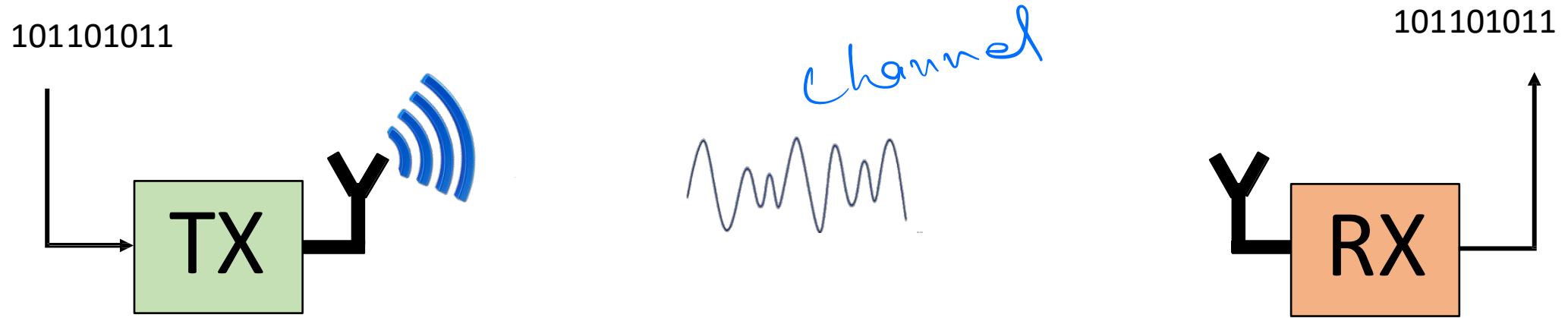
101101011



$$\begin{aligned}
 x(t) &\Rightarrow x(t) \times e^{-j2\pi f_c t} \Rightarrow h(t) * \underline{x(t)e^{-j2\pi f_c t}} \Rightarrow h(t) * x(t)e^{-j2\pi f_c t} \times \underline{e^{j2\pi f_c t}} \\
 &\Rightarrow h(t) * x(t) \\
 &\Rightarrow \underline{y(t) = h(t) * x(t) + v(t)}
 \end{aligned}$$

Assumes TX & RX perfectly synched

Carrier Frequency Offset



$$x(t) \rightarrow x(t) \times e^{-j2\pi f_c t} \rightarrow h(t) * x(t) e^{-j2\pi f_c t} \rightarrow h(t) * x(t) e^{-j2\pi f_c t} \times e^{j2\pi f'_c t}$$

TX & RX are not synched

$$\rightarrow h(t) * x(t) e^{-j2\pi \Delta f_c t}$$

$$\text{CFO: } \Delta f_c = f_c - f'_c$$

$$\rightarrow y(t) = h(t) * x(t) e^{-j2\pi \Delta f_c t} + v(t)$$

Phase changes with time!

cfo compensation

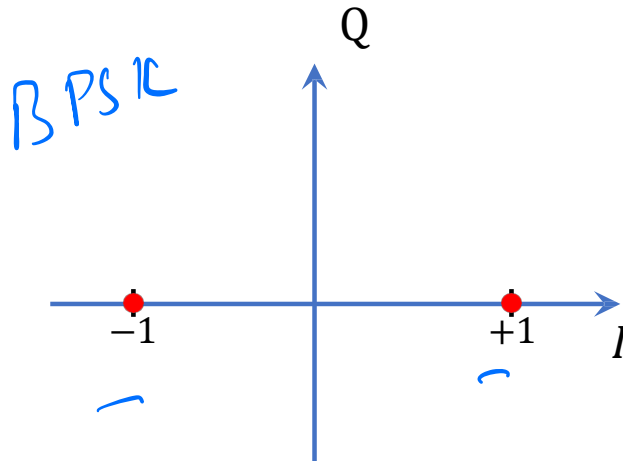
$$\rightarrow y(t) \times e^{+j2\pi \Delta f_c t} = h(t) * x(t)$$

Carrier Frequency Offset

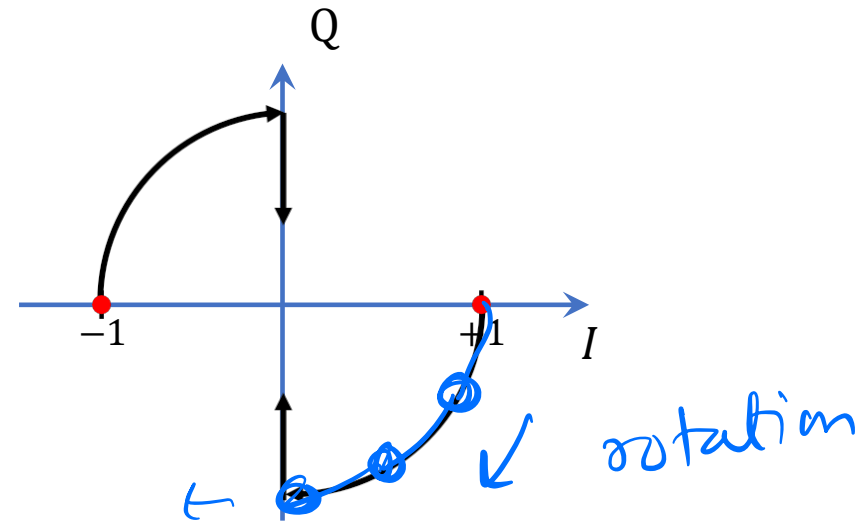
Consider BPSK Modulation.

$0 \rightarrow -1$

$1 \rightarrow +1$



$x(t)$



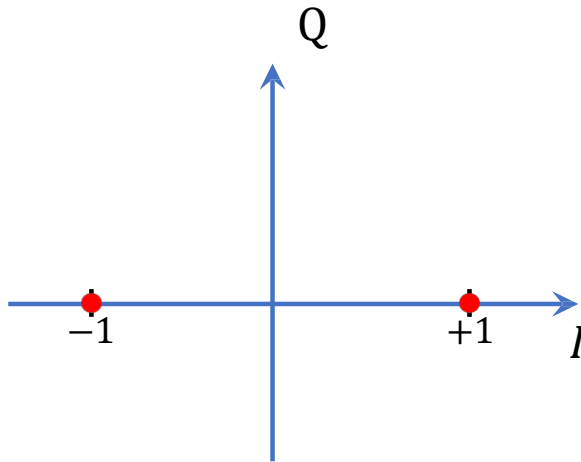
$$h x(t - r) \underbrace{e^{-j2\pi\Delta f_c t}} + v(t)$$

Carrier Frequency Offset

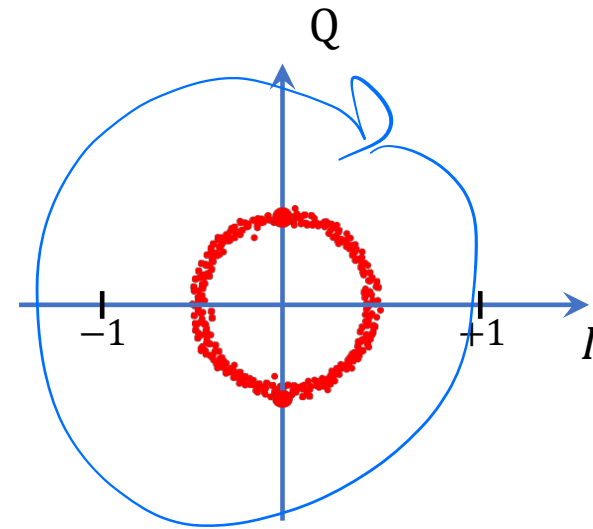
Consider BPSK Modulation.

$0 \rightarrow -1$

$1 \rightarrow +1$



$x(t)$

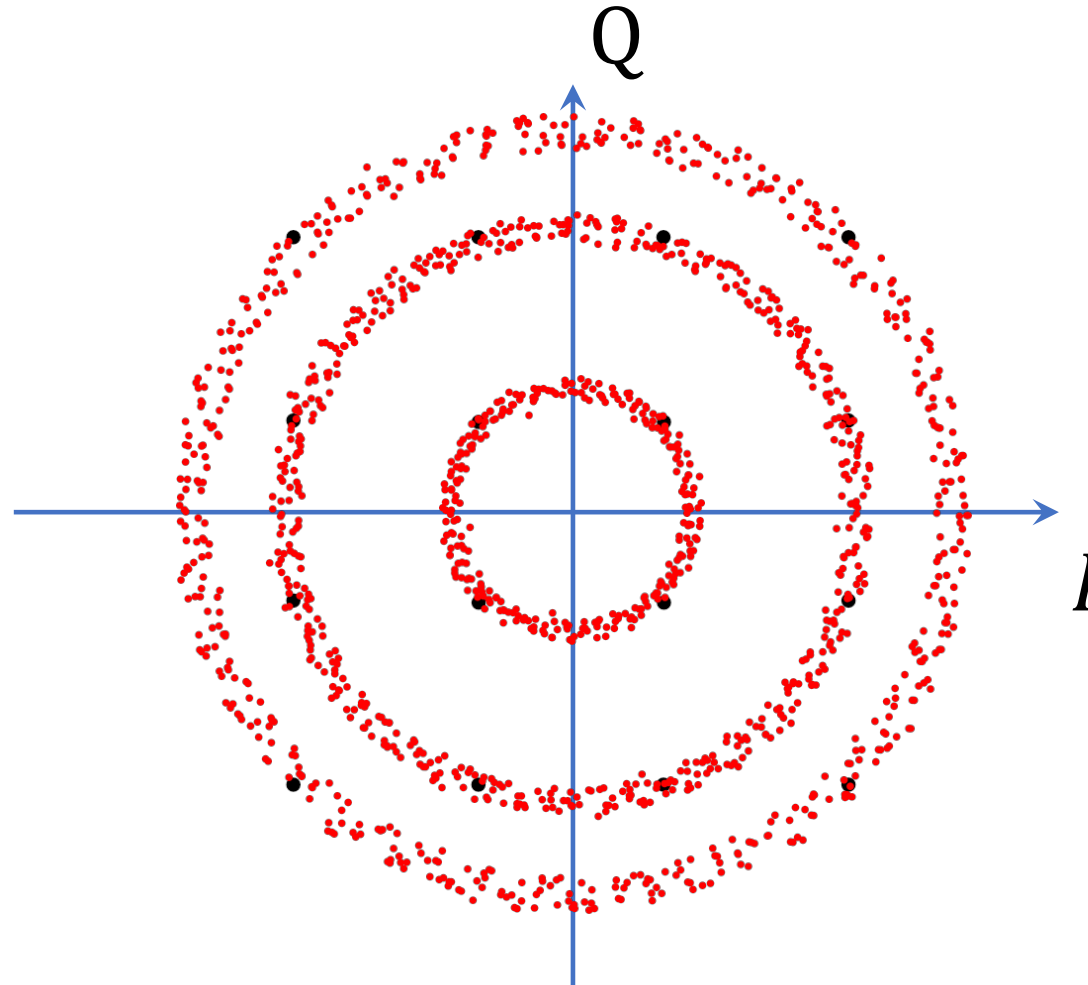


$$h x(t - r) e^{-j2\pi\Delta f_c t} + v(t)$$

Impossible to Decode!

Carrier Frequency Offset

Consider 16 QAM Modulation



Need to estimate and correct CFO to decode!

OFDM CFO Estimation & Correction



- Use Preamble to estimate CFO

$$y_1(t) = x(t)e^{-j2\pi\Delta f_c t}$$

$$y_2(t) = x(t)e^{-j2\pi\Delta f_c (t + T)}$$

Handwritten notes:
 Preamble (next to $y_1(t)$)
 far apart by N samples or NT_s time (next to T)
 Sample at $t = nT_s$
 Symbol has N samples $\rightarrow T = NT_s$

OFDM CFO Estimation & Correction



- Use Preamble to estimate CFO

$\underbrace{LTS_1}$ $y_1[n] = x[n] e^{-j2\pi\Delta f_c n T_s}$

$\downarrow$ same

Sample at $t = nT_s$
 Symbol has N samples $\rightarrow T = NT_s$

$\underbrace{LTS_2}$ $y_2[n] = x[n] e^{-j2\pi\Delta f_c (nT_s + NT_s)}$

- Compute: $A = \sum_{t=1}^N y_1[n] y_2^*[n] = \sum_{t=1}^N x[n] x^*[n] e^{j2\pi\Delta f_c NT_s}$

$$= e^{j2\pi\Delta f_c NT_s} \sum_{t=1}^N |x[n]|^2$$

$$\Rightarrow \Delta f_c = \frac{\angle A}{2\pi NT_s}$$

OFDM CFO Estimation & Correction



- Use Preamble to estimate CFO

$$y_1[n] = x[n]e^{-j2\pi\Delta f_c nT_s}$$

$$y_2[n] = x[n]e^{-j2\pi\Delta f_c (nT_s + NT_s)}$$

entire package
Preamble
(SIFS + LTR)
+ data + CP
100 ofdm
symbols

1 packet

- Compute:
$$A = \sum_{t=1}^N y_1[n]y_2^*[n] \quad \rightarrow \quad \Delta f_c = \frac{\angle A}{2\pi nT_s}$$

- Correct CFO: $y[n] \times e^{j2\pi\Delta f_c nT_s}$

entire packet including data & preamble
Assume Δf_c or CFO is constant for one OFDM packet

OFDM Channel Estimation



- Use Preamble to estimate the channel

$$y(t) = h(t) * x(t) \leftrightarrow Y(f) = H(f) X(f)$$

- Send $X(f)$: $-1, +1, -1, -1, -1, +1, \dots$

- Receive: $-H(1), H(2), -H(3), -H(4), -H(5), H(6), \dots$

- Estimate: $\tilde{H}(f) = \frac{Y(f)}{X(f)}$

- Use two preambles to average noise:

$\Rightarrow$ channel equalization for data

$$x_{\text{data}} = y_{\text{data}}(f) / \tilde{H}(f)$$

Handwritten notes:

$$Y \rightarrow Y'$$

$$= y e^{-j2\pi k 4 T_s}$$

gets absorbed in $\tilde{H}$

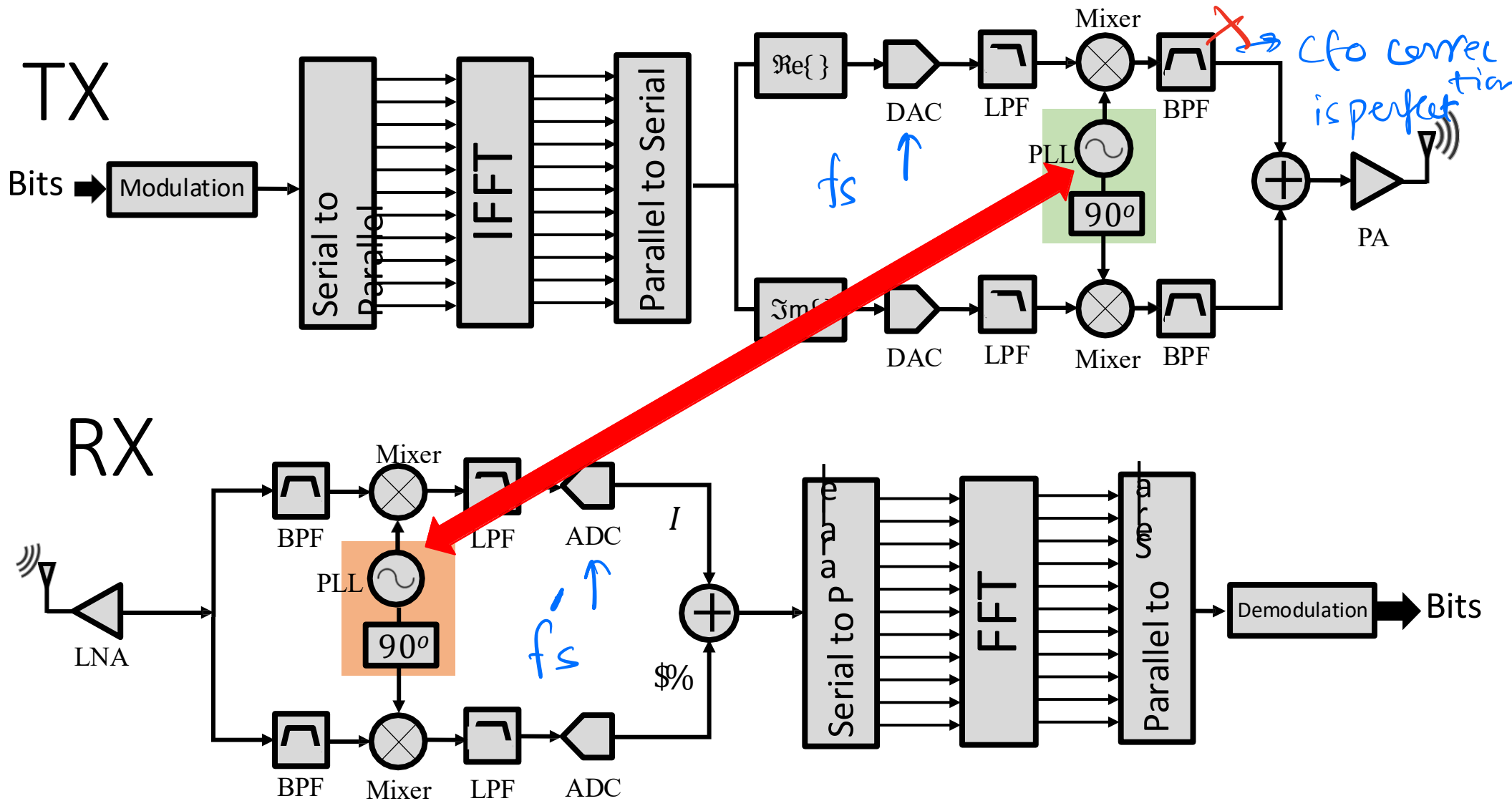
LTS_1 and LTS_2 are indicated with arrows pointing to the terms in the equation above.

$$\tilde{H}(f) = \frac{Y_1(f) + Y_2(f)}{2 X(f)}$$

OFDM: Orthogonal Frequency Division

Multiplexing $\times \rightarrow$

Df_c or Cf_c
is constant
for entire OFDM
packet



Phase Tracking

So Far: Estimated and Corrected For Coarse Value of CFO

- Residual CFO:

$$y(t) = h(t) * x(t) e^{j2\pi\Delta f_c t} + v(t)$$

$$\Delta f_c = df_c + \delta f_c$$

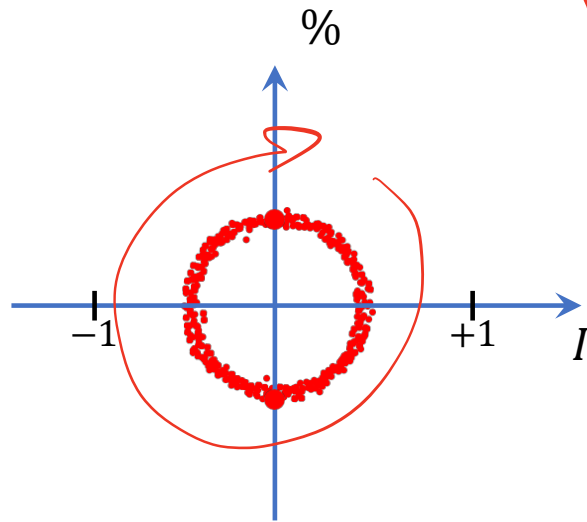
The diagram illustrates the decomposition of the total carrier frequency offset (CFO), Δf_c , into two parts: a coarse CFO, df_c , and a residual CFO, δf_c . A blue checkmark is placed above the equation. A red arrow points from the underlined text 'Coarse CFO' to df_c . Another red arrow points from the underlined text 'Residual CFO' to δf_c , which is circled in red. A red bracket groups the two terms on the right side of the equation.

We estimated and corrected for coarse CFO!

Even small residual can accumulate over time
to create large phase: $e^{-j2\pi\delta f_c t}$

Need to track the phase

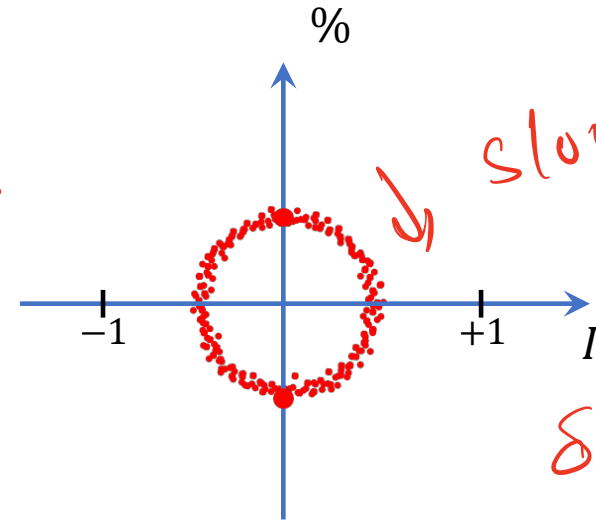
Phase Tracking



full CFO
 Δf_c is large

$$h x(t - r) e^{j2\pi \Delta f_c t} + v(t)$$

Without Course CFO Correction



slower rotation

δf_c is small

$$h x(t - r) e^{j2\pi \delta f_c t} + v(t)$$

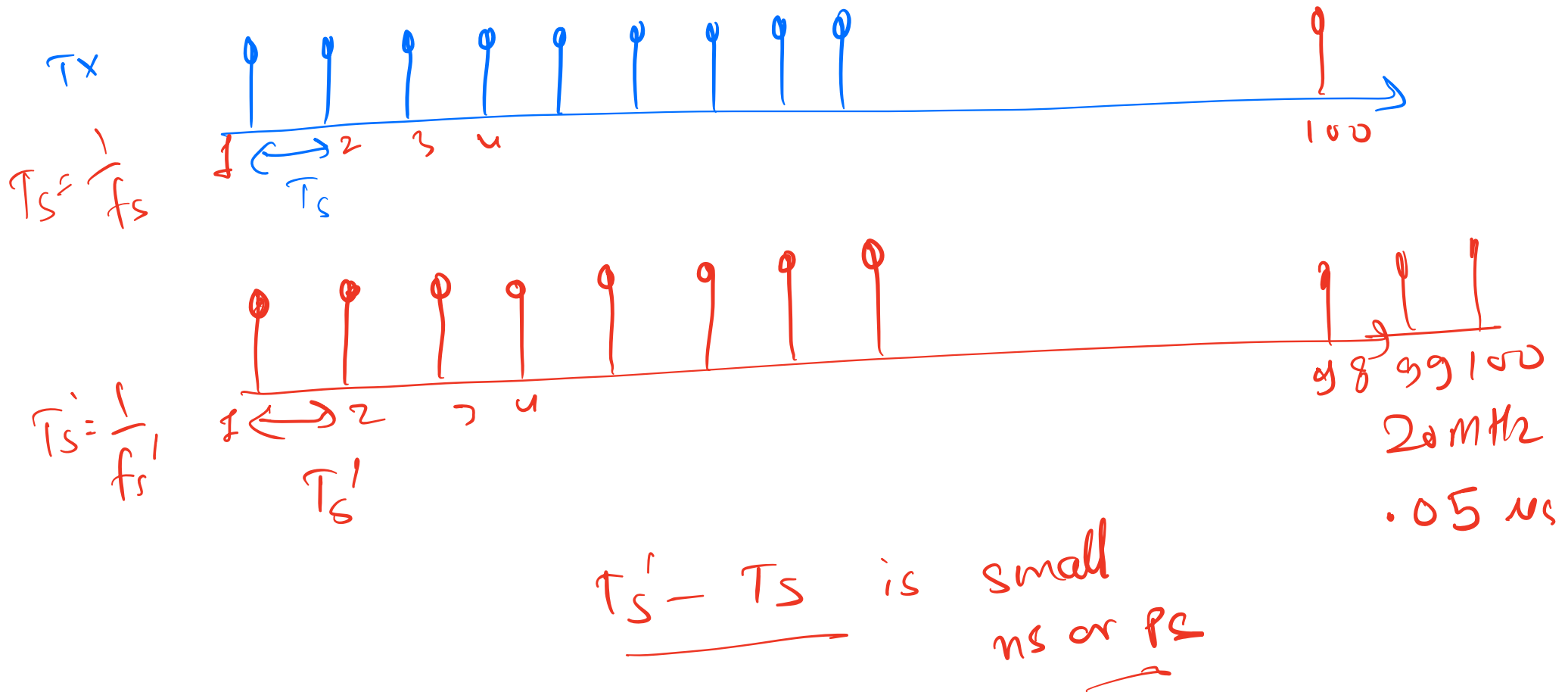
Without Residual CFO Correction

$$2\pi \delta f_c t_1 - 2\pi \delta f_c t_2$$

$$(t_1 - t_2)$$

Phase Tracking

- Residual CFO (Carrier Frequency Offset)
- Residual SFO (Sampling Frequency Offset)



Phase Tracking

- Residual CFO (Carrier Frequency Offset)
- Residual SFO (Sampling Frequency Offset)

When we sample the signal there is a residual sampling offset: $n\delta T_s$

$$\delta T_s = T_s - T_s'$$

Phase Tracking

- Residual CFO (Carrier Frequency Offset)
- Residual SFO (Sampling Frequency Offset)

When we sample the signal there is a residual sampling offset: $n\delta T_s$

$$y[n] = x[n + n\delta T_s] e^{-j2\pi\delta f_c n T_s}$$

$\xrightarrow{\text{SFO}}$ $\xleftarrow{\text{residual CFO}}$

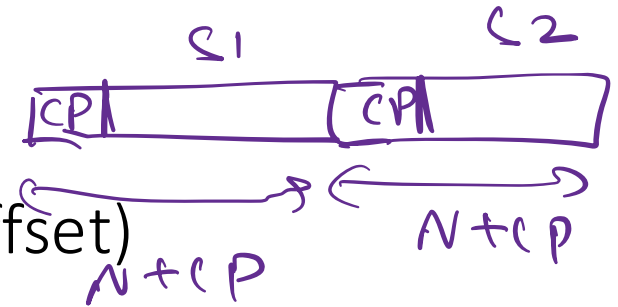
treat as constant

$$Y[f] = X[f] e^{j \frac{2\pi f n \delta T_s}{N} - 2\pi\delta f_c n T_s}$$

$\xrightarrow{\text{SFO}}$ $\xleftarrow{\text{residual CFO}}$

$\uparrow$ freq independent
 $\uparrow$ is freq dependent $\Rightarrow$ linear phase with freq

Phase Tracking



- Residual CFO (Carrier Frequency Offset)
- Residual SFO (Sampling Frequency Offset)

Pilot X

$$\rightarrow \underline{Y_1[f]} = \underline{X_1[f]} e^{j \frac{2\pi f n \delta T_s}{N} - 2\pi \delta f_c n T_s}$$

same as before
 $n \rightarrow n + N + CP$

$$\rightarrow \underline{Y_2[f]} = \underline{X_2[f]} e^{j \frac{2\pi f (n + N + CP) \delta T_s}{N} - 2\pi \delta f_c (n + N + CP) T_s}$$

Phase accumulation: $\Delta \phi = 2\pi f \frac{(N + CP) \delta T_s}{N} - 2\pi \delta f_c (N + CP) T_s$

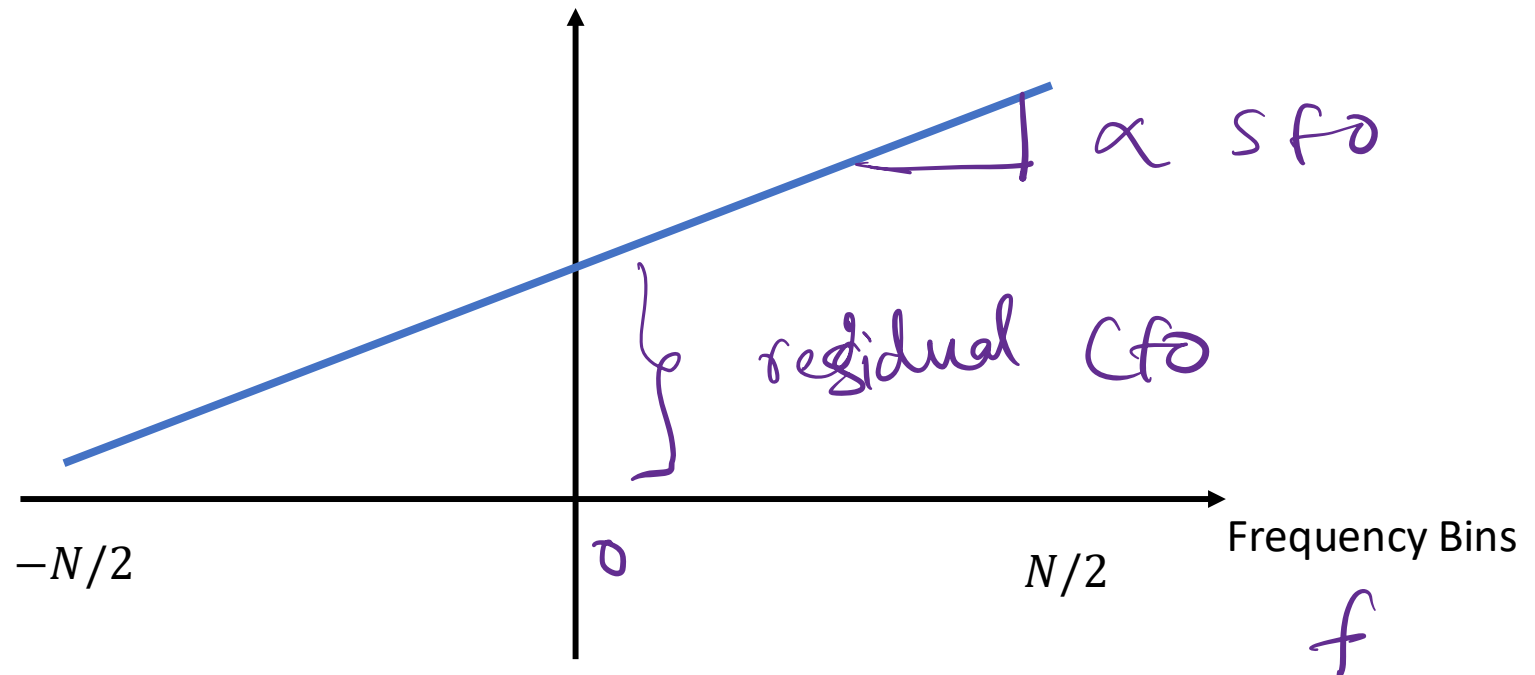
SFO

residual CFO

Phase Tracking

- Residual CFO (Carrier Frequency Offset)
- Residual SFO (Sampling Frequency Offset)

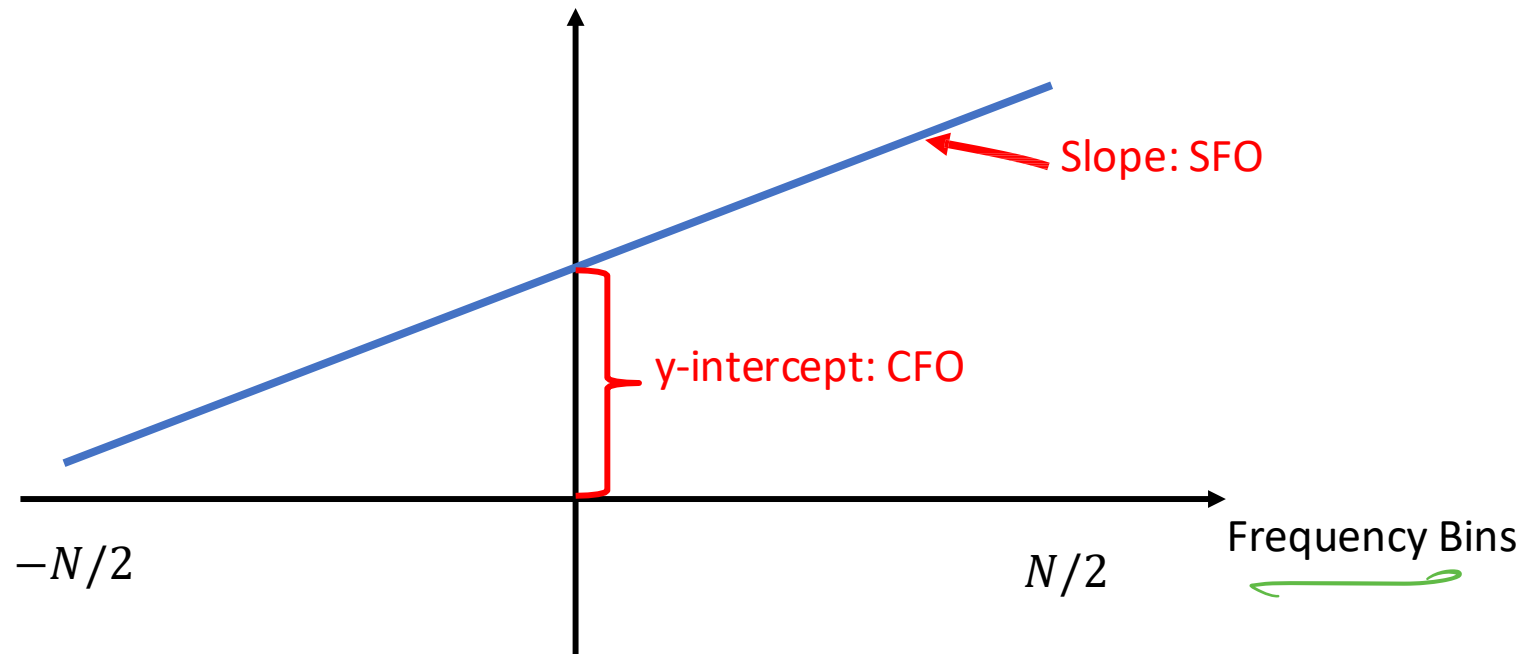
- Phase accumulation: $\Delta\phi = 2\pi f \frac{(N + CP)\delta T_s}{N} - 2\pi\delta f_c(N + CP)T_s$



Phase Tracking

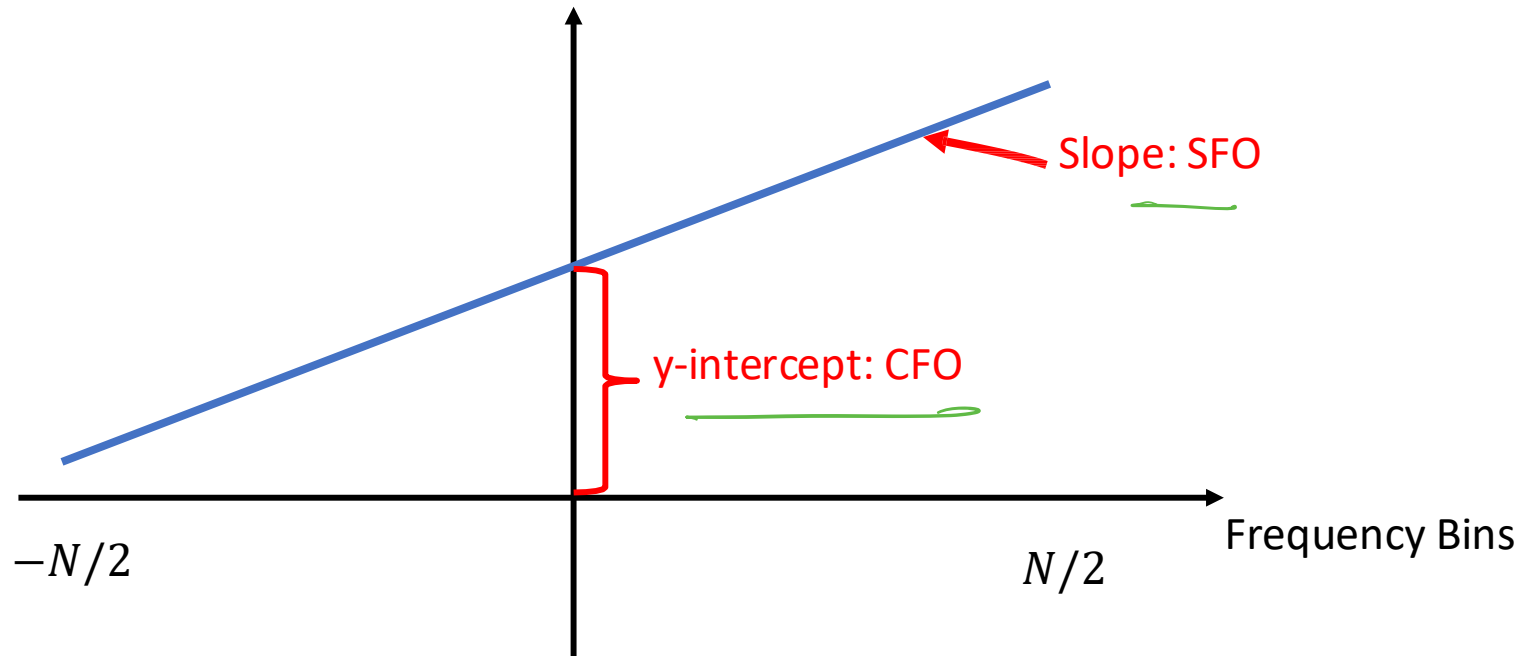
- Residual CFO (Carrier Frequency Offset)
- Residual SFO (Sampling Frequency Offset)

- Phase accumulation: $\Delta\phi = 2\pi f \frac{(N + CP)\delta T_s}{N} - 2\pi\delta f_c(N + CP)T_s$



OFDM Phase Tracking

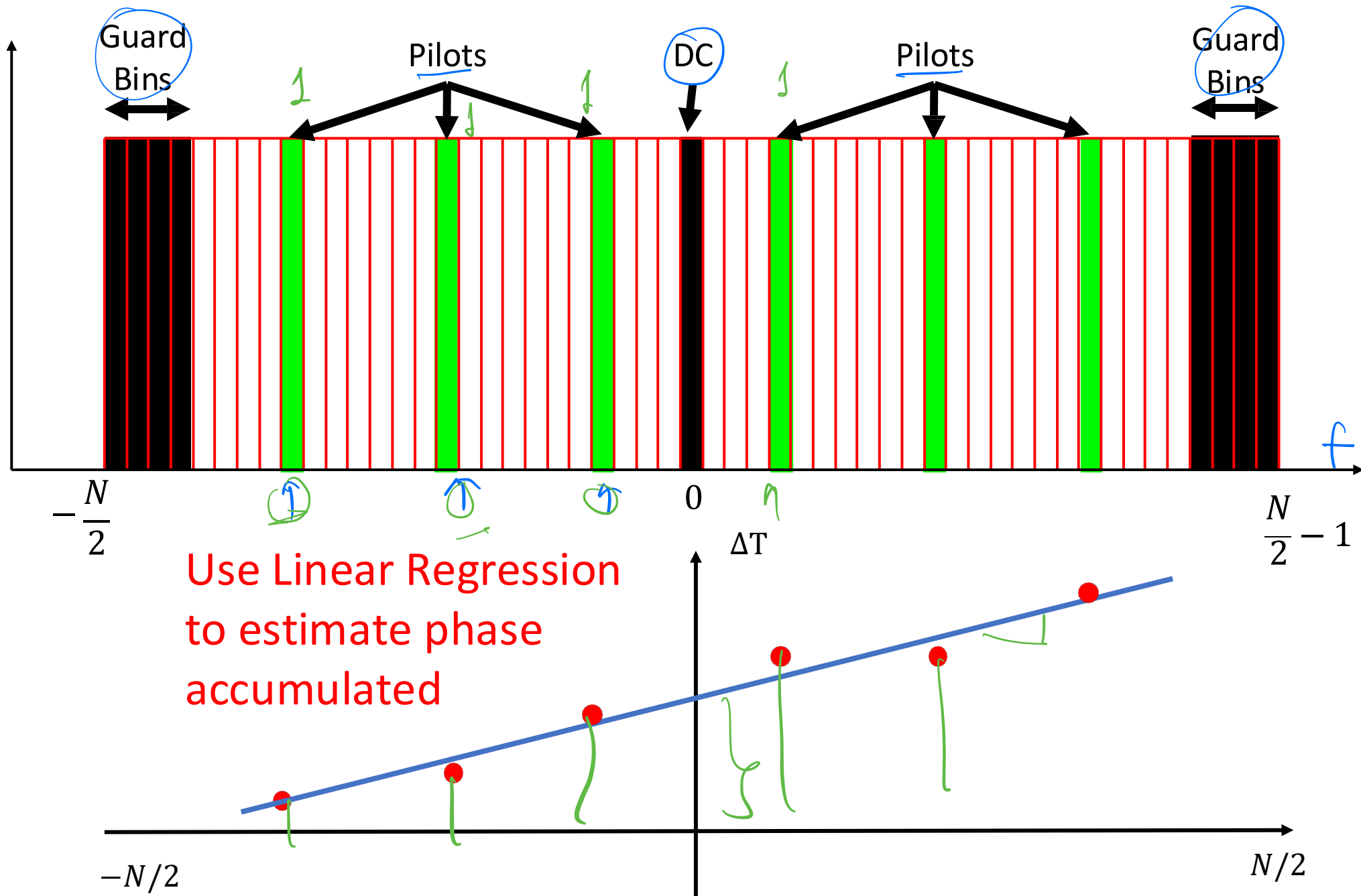
- Phase accumulation: $\Delta\phi = 2\pi f \frac{(N + CP)\delta T_s}{N} - 2\pi\delta f_c(N + CP)T_s$



- Sufficient to estimate slope & y-intercept to know the phase accumulated for all subcarriers.
- Use only few subcarriers as pilots & send known bits in them.

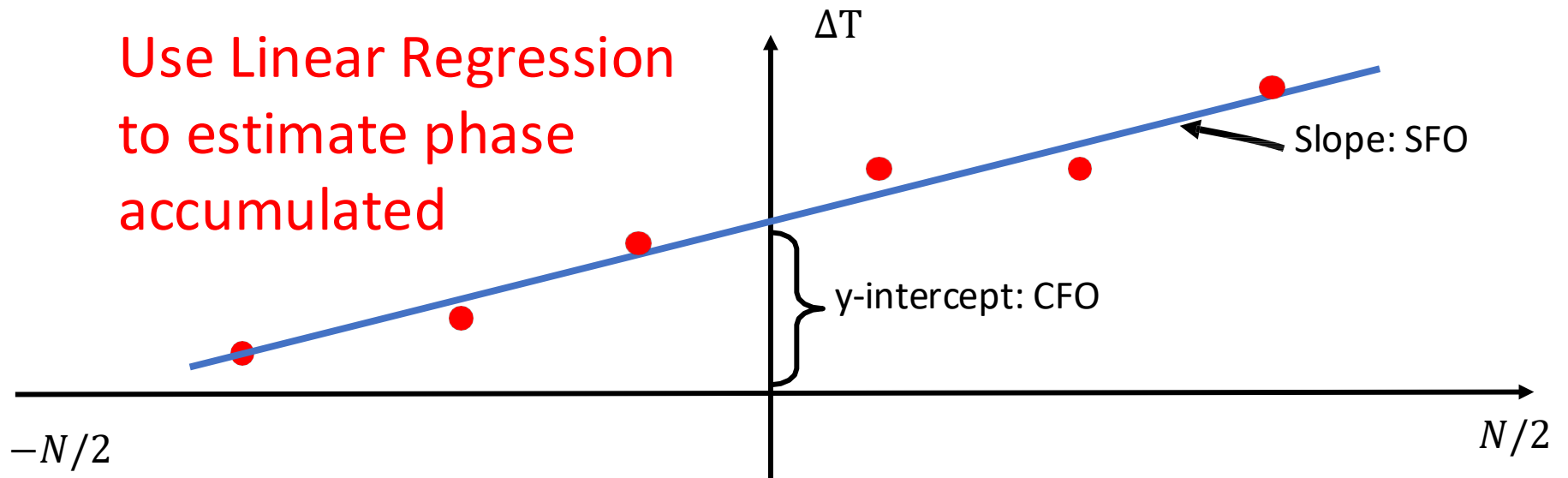
OFDM Phase Tracking

$s_1, s_2, \dots$



OFDM Phase Tracking

Use Linear Regression
to estimate phase
accumulated

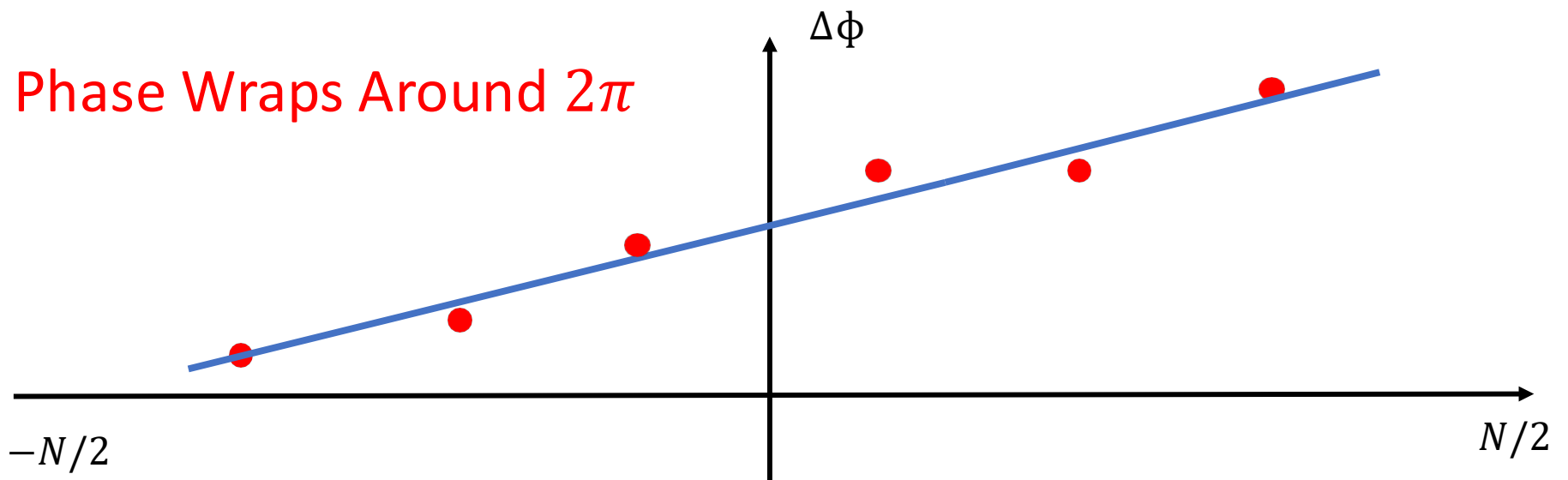


$$\text{Slope} = 2\pi \frac{(N + CP)\delta T_s}{N}$$

$$Y - \text{intercept} = -2\pi\delta f_c(N + CP)T_s$$

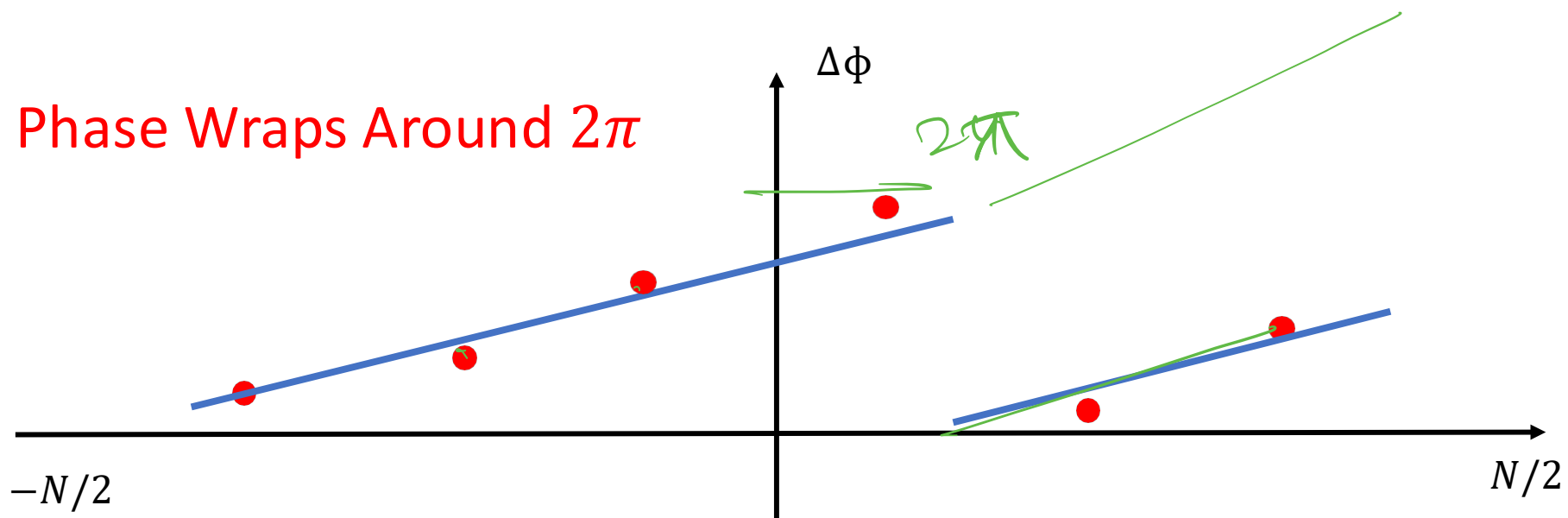
$$\Delta\phi = 2\pi f \frac{(N + CP)\delta T_s}{N} - 2\pi\delta f_c(N + CP)T_s$$

Be Careful!

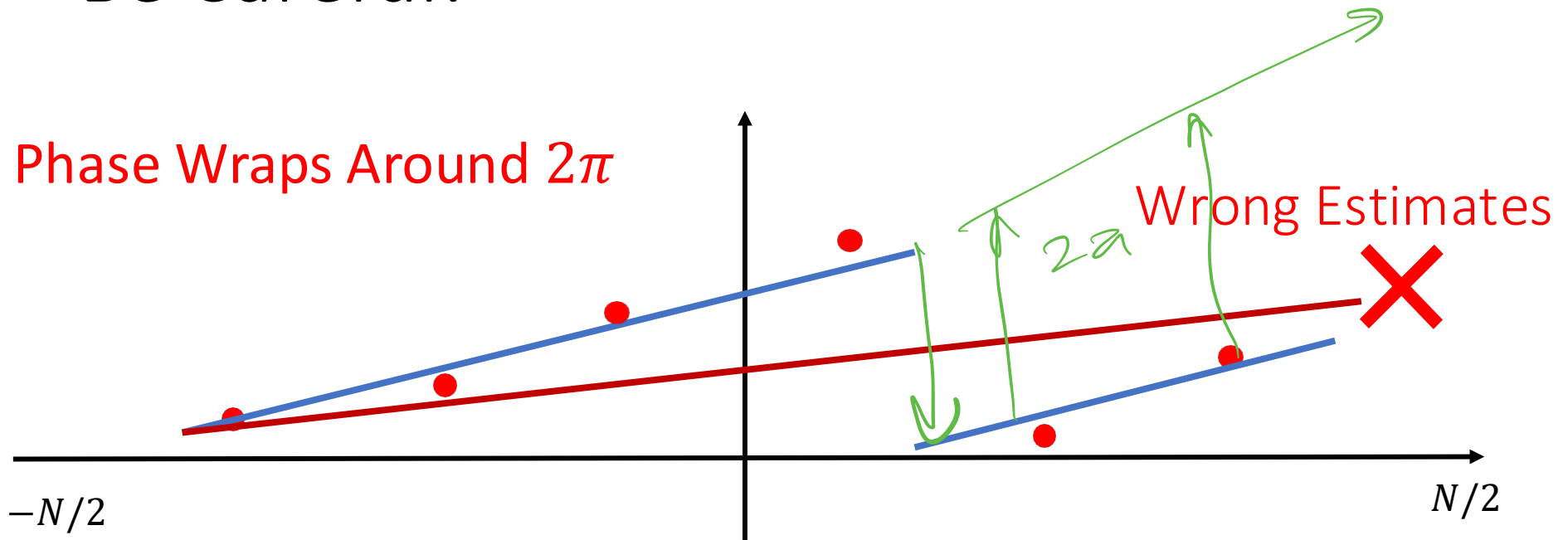


Be Careful!

Phase Wraps Around 2π



Be Careful!

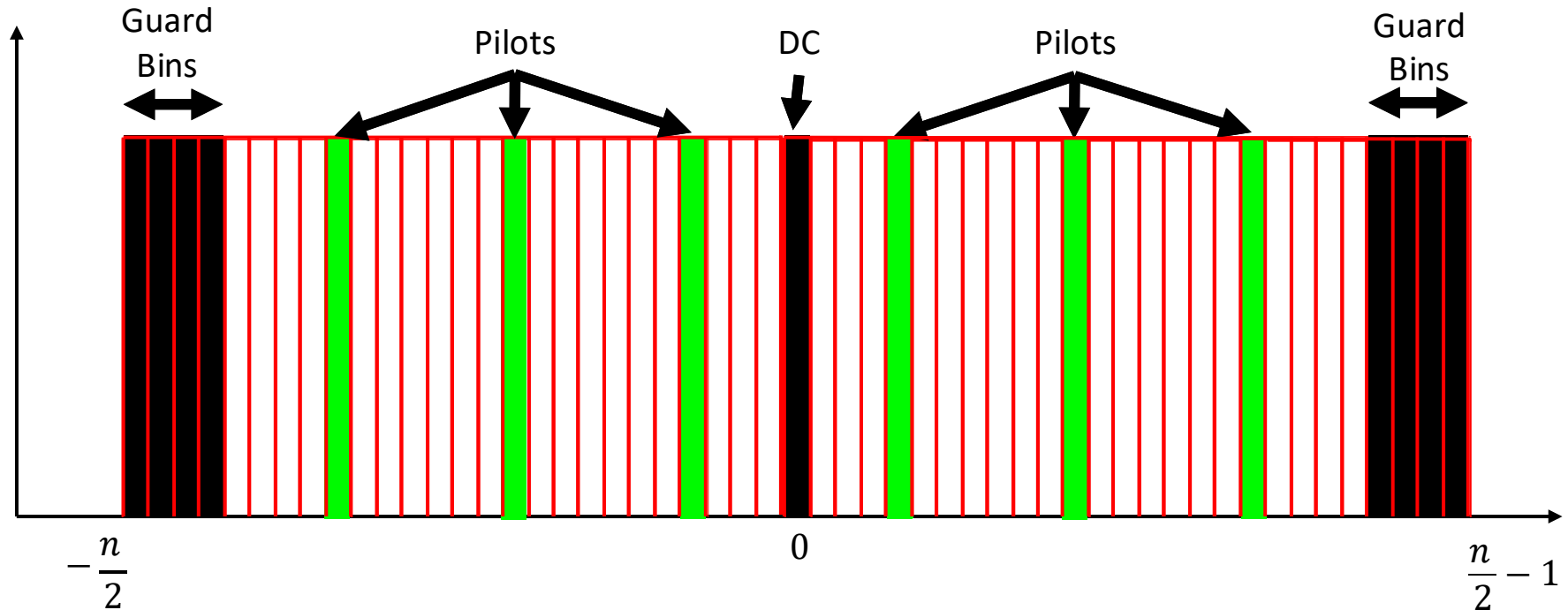


Need to unwrap phase mod 2π first! Use `unwrap()` function in MATLAB

When estimating slope and y-intercept, equation assumes subcarriers index f is between $-n/2$ and $n/2 - 1$

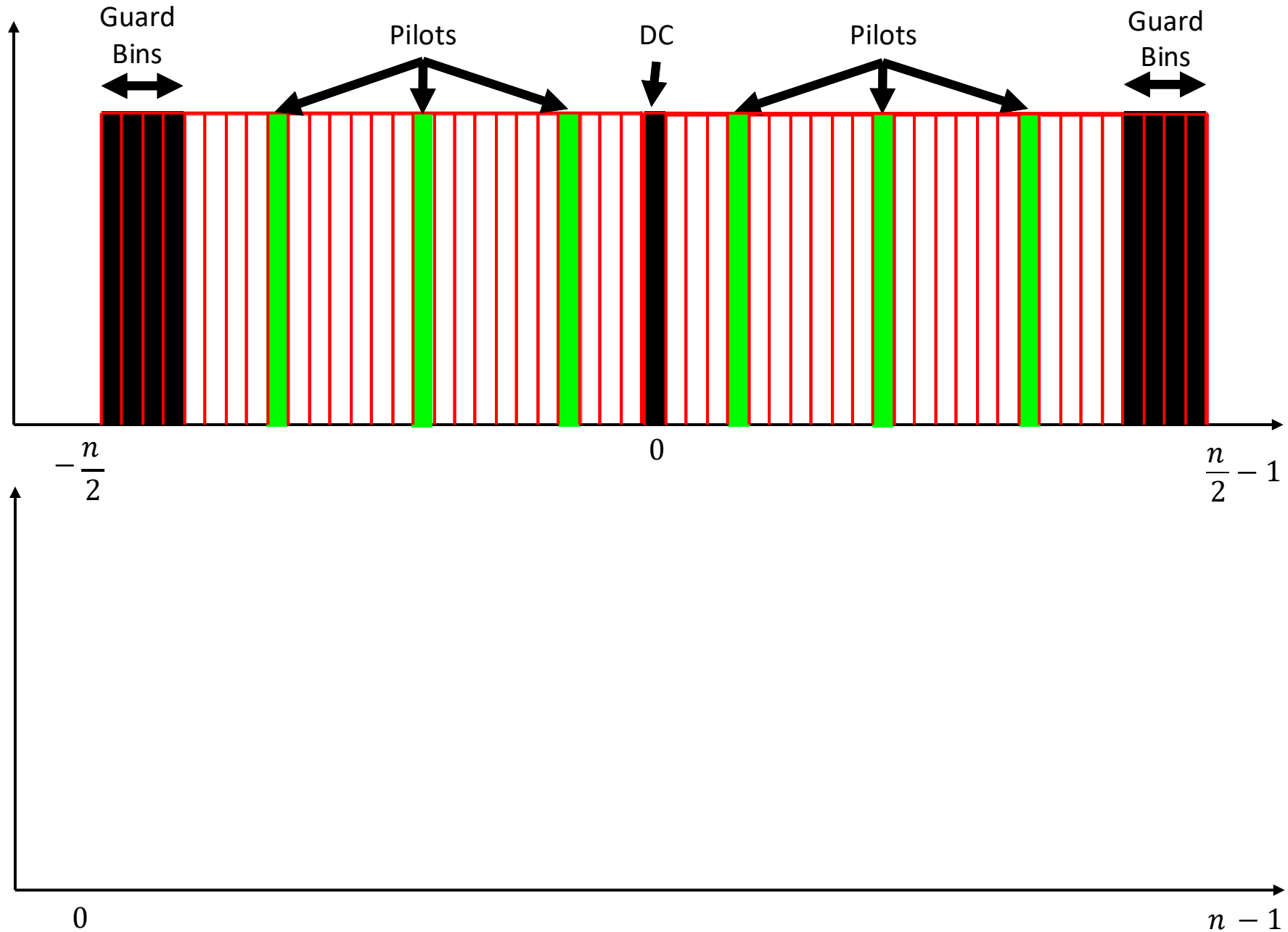
Be Careful!

When estimating slope and y-intercept, equation assumes subcarriers index f is between $-\frac{n}{2}$ and $\frac{n}{2} - 1$



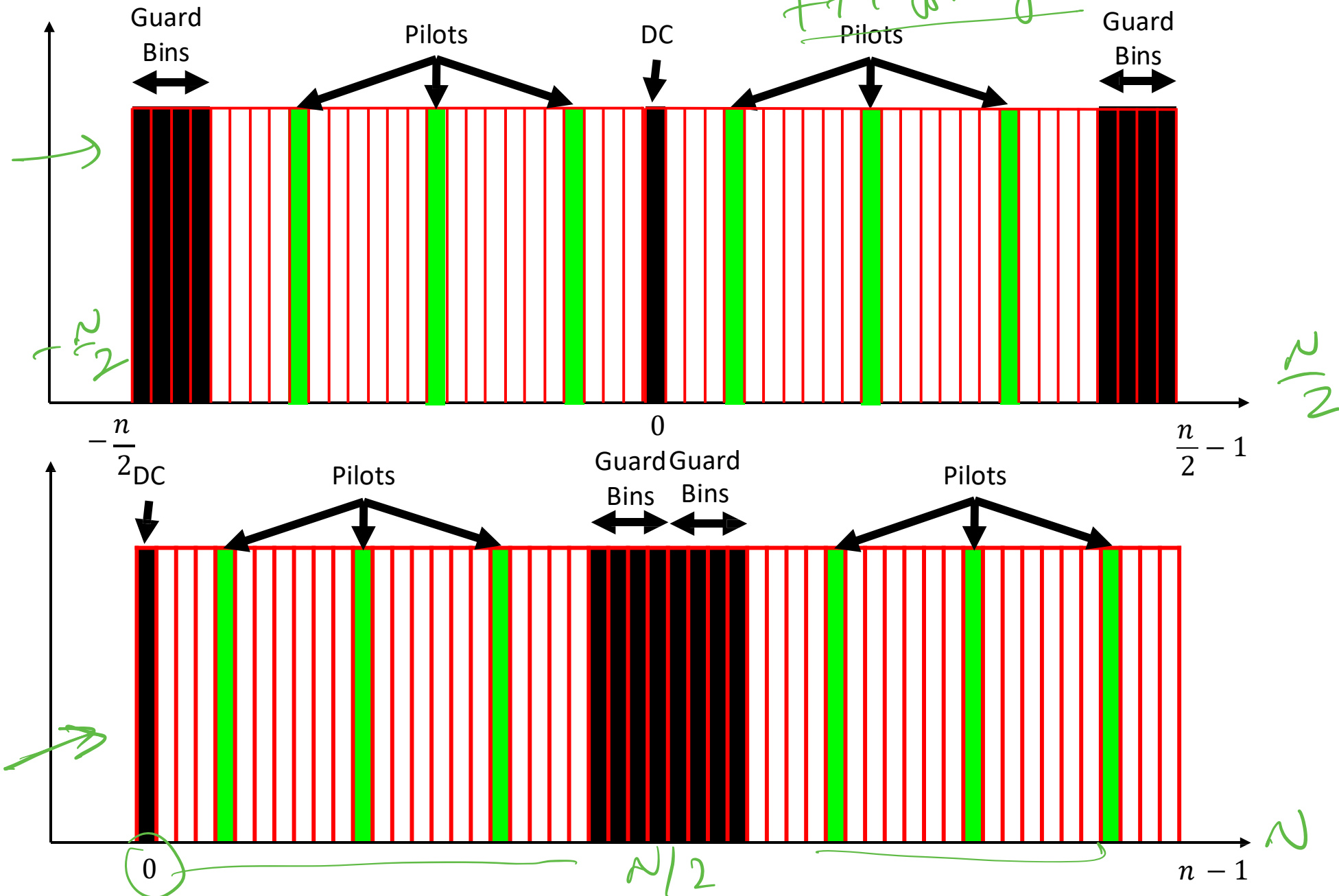
Be Careful!

When estimating slope and y-intercept, equation assumes subcarriers index f is between $-\frac{n}{2}$ and $\frac{n}{2} - 1$



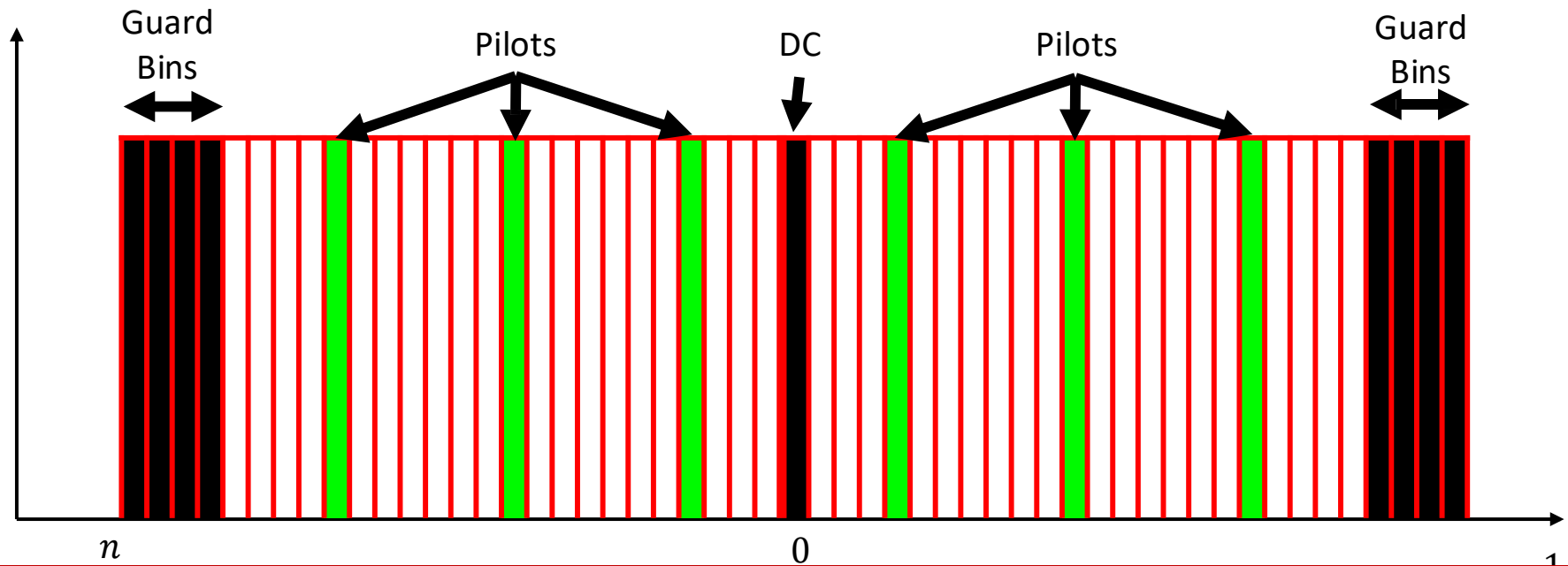
Be Careful!

When estimating slope and y-intercept, equation assumes subcarriers index + is between $-n/2$ and $n/2 - 1$

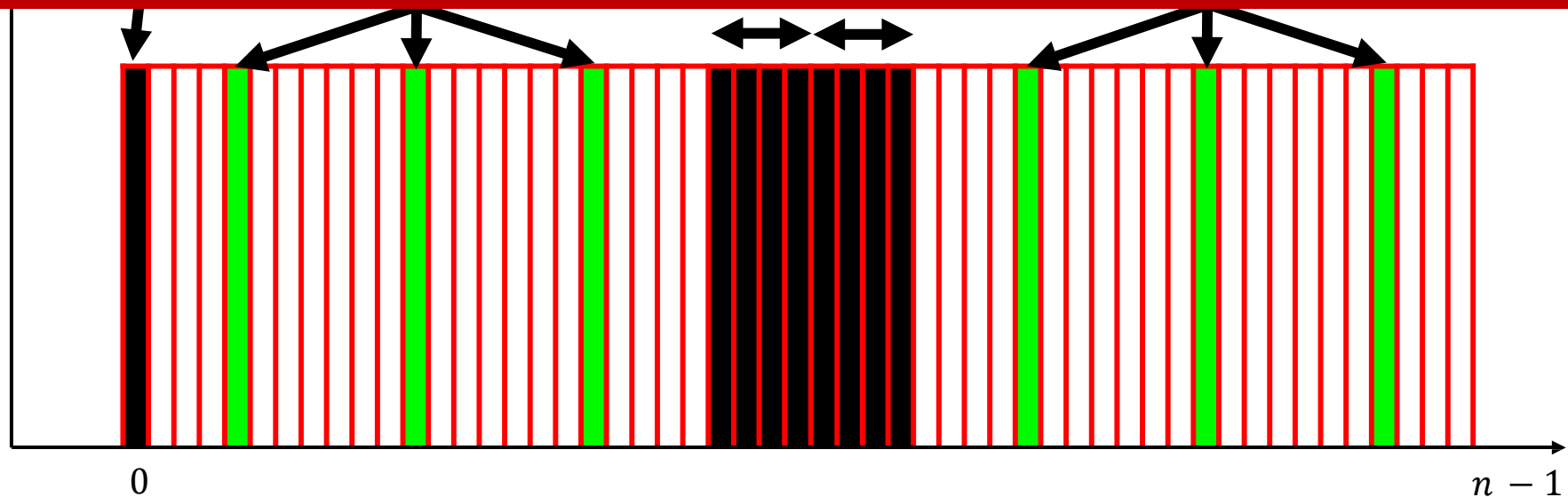


Be Careful!

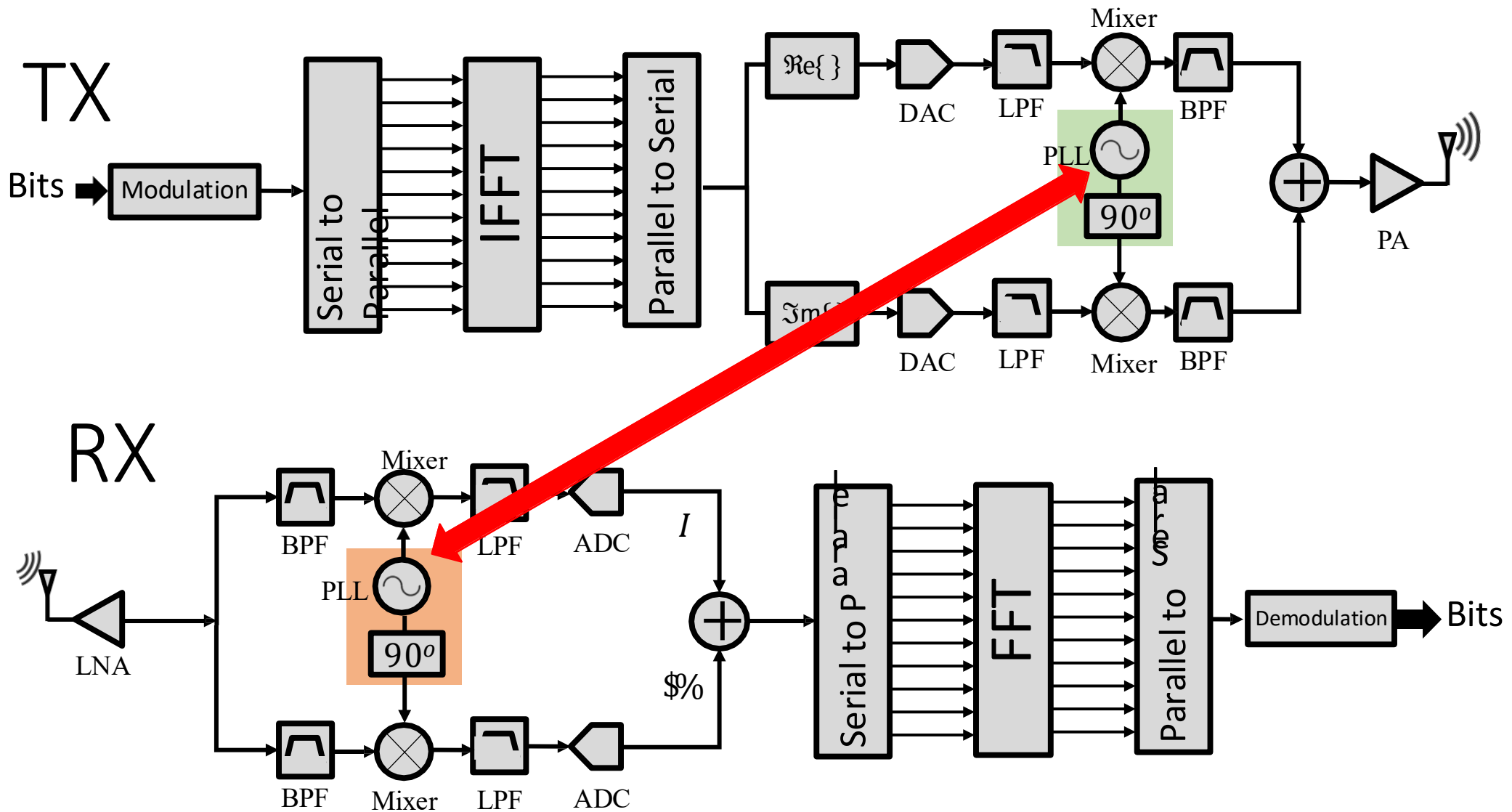
When estimating slope and y-intercept, equation assumes subcarriers index + is between $-n/2$ and $n/2 - 1$



Need to `fftshift()` function in MATLAB + provided functions to convert indexing.



OFDM: Orthogonal Frequency Division Multiplexing



OFDM: Putting it Together

At TX:

- Create preamble symbol from training sequence (Uses BPSK)
- Repeat preamble symbol:
 - ~10 times STS for coarse packet detection
 - 2 times LTS for fine packet detection
 - 2 times LTS for CFO estimation
 - 2 times LTS for channel estimation
 - Add CP for LTS preamble
- Create data symbol from:
 - Data bits (Uses BPSK, QPSK, M-QAM)
 - Pilot bits (Uses BPSK)
- Add cyclic prefix to data symbols.

OFDM: Putting it Together

At RX (HW 4):

- Detect beginning of packet.
- Estimate & correct for CFO.
- Jump $\approx CP - 4$ samples into symbol to avoid ISI
- Estimate the channel.
- For each subsequent data symbol:
 - Remove CP
 - Take FFT of Size N
 - Correct for channel by dividing with $\tilde{H}(f)$
 - Use linear regression to estimate residual CFO and SFO with pilots
 - Estimate accumulated phase $\Delta\phi_f$ for each frequency bin
 - Correct for residual CFO and SFO
 - Decode Bits